

$$\ln\left(\frac{WTP_{13}}{WTP_x}\right) = \beta \times (13 - x)$$

Exponentiating and rearranging terms,

$$WTP_x = WTP_{13} \times e^{-\beta \cdot (13-x)}$$

There is uncertainty surrounding the exact values of WTP13; x, and β , and this uncertainty can be incorporated in the equation, if you request that the analysis be carried out in “uncertainty mode.” The distribution of WTP to avoid a severe case of chronic bronchitis, WTP13, is based on the distribution of WTP responses in the Viscusi et al. (1991) study. The distribution of x, the severity level of an average case of pollution-related chronic bronchitis, is modeled as a triangular distribution centered at 6.5, with endpoints at 1.0 and 12.0. And the distribution of β is normal with mean = 0.18 and std. dev.= 0.0669 (the estimate of b and standard error reported in Krupnick and Cropper, 1992).

In uncertainty mode, BenMAP uses a Monte Carlo approach. On each Monte Carlo iteration, random draws for these three variables are made, and the resulting WTP_x is calculated from the equation above. Because this function is non-linear, the expected value of WTP for a pollution-related case of CB cannot be obtained by using the expected values of the three uncertain inputs in the function (doing that will substantially understate mean WTP). A Monte Carlo analysis suggests, however, that the mean WTP to avoid a case of pollution-related chronic bronchitis is about \$470,000 (2015\$), but not adjusted for the growth of income). Therefore, if you request that the analysis be carried out in “point estimate” mode, that is the unit value that is used.

I.3.1.2 Alternative Cost of Illness Estimates

Cost of illness estimates for chronic bronchitis were derived from estimates of annual medical costs and annual lost earnings by Cropper and Krupnick (1999). This study estimated annual lost earnings resulting from chronic bronchitis as a function of age at onset of the illness, for the following age categories: 25-43, 35-44, 45-54, and 55-65 (see Cropper and Krupnick, Table 8). Annual medical expenses were estimated for 10-years age groups (0-9, 10-19, 20-29, ..., 80-89). We derived estimates of the present discounted value of the stream of medical and opportunity costs for people whose age of onset is 30, 40, 50, 60, 70, and 80. Medical costs (which are in 1977\$ in the Cropper and Krupnick study) were inflated to 2015\$ using the CPI-U for medical care; lost earnings (opportunity costs) were inflated to 2015\$ using the Employment Cost Index for Wages and Salaries. Life expectancies were assumed to be unaffected by the illness. For example, an individual at age 70 has a life expectancy of 14.3 more years, and we assumed that someone whose age of onset of chronic bronchitis is 70 will also live for 14.3 more years. (Source of life expectancies: National Center for Health Statistics, 1999, Table 5.) We also assumed that opportunity costs at ages 66 and over were zero. Present discounted values were calculated using three and seven percent discount rates.

For each of the two discount rates, there are three cost of illness unit values for chronic bronchitis available in BenMAP, for the following age categories: 27-44, 45-64, and 65+. These are the age categories that were used in the epidemiological study that estimated a concentration-response function for chronic bronchitis (Abbey et al., 1995b). The estimate for the 27-44 age group is an average of the present discounted values calculated for ages 30 and 40; the estimate for the 45-64 age category is an average of the present discounted values calculated for ages 50 and 60; and the estimate for the 65+ age category is an average of the present discounted values calculated for ages 70 and 80. The suite of unit values available for use in BenMAP is shown in Table I-3.

Table I-3. Additional Unit Values Available for Chronic Bronchitis

Basis for Estimate	Age of Onset		Present Discounted Value of Medical Costs	Present Discounted Value of Opportunity Costs	Unit Value	Distribution
	Min	Max				
WTP: average severity	30	99	N/A	N/A	\$468,641	Custom
COI: med costs + wage loss, 3% DR	27	44	\$32,478	\$199,479	\$231,947	None
	45	64	\$40,699	\$111,959	\$152,658	None
	65	99	\$18,993	\$0	\$18,993	None
COI: med costs + wage loss, 7% DR	27	44	\$13,509	\$118,460	\$131,969	None
	45	64	\$24,651	\$87,732	\$112,383	None
	65	99	\$15,468	\$0	\$15,468	None

I.3.2 Chronic Bronchitis Reversals

The unit value for chronic bronchitis reversals assumes that this is chronic bronchitis with a severity level of 1. The method for generating a distribution of unit values in BenMAP is therefore the same as the WTP-based unit value method for chronic bronchitis (see above), with $x = 1$. The mean of this distribution is \$206,765.

I.3.3 Chronic Asthma

Two studies have estimated WTP to avoid chronic asthma in adults. Blumenschein and Johannesson (1998) used two different contingent valuation (CV) methods, the dichotomous choice method and a bidding game, to estimate mean willingness to pay for a cure for asthma. The mean WTP elicited from the bidding game was \$189 per month, or \$2,268 per year (in 1996\$). The mean WTP elicited from the dichotomous choice approach was \$343 per month, or \$4,116 per year (in 1996\$). Using \$2,268 per year, a three percent discount rate, and 1997 life expectancies for males in the United States (National Center for Health Statistics, 1999, Table 5), the present discounted value of the stream of annual WTPs is \$65,568 (in 2015\$).

O'Connor and Blomquist (1997) estimated WTP to avoid chronic asthma from estimates of risk-risk tradeoffs. Combining the risk-risk tradeoffs with a statistical value of life, the annual value of avoiding asthma can be derived. Assuming a value of a statistical life of

\$6 million, they derived an annual WTP to avoid asthma of \$1500 (O'Connor and Blomquist, 1997, p. 677). For a value of a statistical life of \$5,894,400 (in 1997 \$), the corresponding implied annual value of avoiding chronic asthma, based on O'Connor and Blomquist would be \$1,474. Assuming a three percent discount rate and 1997 life expectancies for males in the United States, the present discounted value of the stream of annual WTPs would be \$41,646 (in 2015\$). A unit value, based on a three percent discount rate, is the average of the two estimates, or \$53,607. Following the method used for the §812 Prospective analysis, the uncertainty surrounding the WTP to avoid a case of chronic asthma among adult males was characterized by a triangular distribution on the range determined by the two study-specific WTP estimates. A second unit value, using a seven percent discount rate, is also available for use in BenMAP. The method used to derive this unit value is the same as that described above for the three percent discount rate unit value. The unit values available for use in BenMAP are summarized in Table I-4 below.

Table I-4. Additional Unit Values Available for Chronic Asthma

Basis for Estimate	Age Range		Unit Value	Distribution of Unit Value	Parameters of Distribution	
	Min	Max			P1	P2
WTP: 3% DR (Discount Rate)	27	99	\$53,607	Triangular	\$41,646	\$65,568
WTP: 7% DR	27	99	\$34,901	Triangular	\$27,114	\$42,689

I.4 Acute Symptoms and Illness Not Requiring Hospitalization

See Appendix H, Section H.3 for a general explanation of acute symptoms and illness not requiring hospitalization. Table I-5 summarizes unit values for acute bronchitis in children, acute respiratory symptoms (minor restricted activity days), any of 19 respiratory symptoms, upper respiratory symptoms, lower respiratory symptoms, and work loss days (WLDs)

Table I-5. Additional Unit Values Available for Acute Symptoms and Illnesses (in 2015 \$)

Health Endpoint	Basis for Estimate *	Age Range		Unit Value	Distribution of Unit Value	Parameters of Distribution	
		Min	Max			P1	P2
Acute Bronchitis	WTP: 1 day illness, CV studies	0	17	\$82	Uniform	24.10	139.16
	WTP: 6 day illness, CV studies	0	17	\$490	Uniform	144.60	834.98
	WTP: 28 symptom-days, Dickie and Ulery	0	17	\$529	Lognormal	6.27	0.0907
Minor Restricted Activity Days	WTP: 3 symptoms 1 day, Dickie and Ulery (2002)	18	99	\$138	Lognormal	4.93	0.0649
Lower Respiratory Symptoms	WTP: 1 day, CV studies	0	17	\$21	Uniform	9.56	33.68
	WTP: 2 symptoms 1 day, Dickie and Ulery (2002)	0	17	\$264	Lognormal	5.58	0.07048

Health Endpoint	Basis for Estimate *	Age Range		Unit Value	Distribution of Unit Value	Parameters of Distribution	
		Min	Max			P1	P2
	WTP: 2x1 day, CV studies	0	17	\$42	Uniform	19.12	67.36
Upper Respiratory Symptoms	WTP: 1 day, CV studies	0	17	\$34	Uniform	12.29	59.34
	WTP: 2 symptoms 1 day, Dickie and Ulery (2002)	0	17	\$264	Lognormal	5.58	0.07048
	WTP: 2x1 day, CV studies	0	17	\$68	Uniform	24.58	118.68
Any of 19 Respiratory Symptoms	WTP: 1 day illness, CV studies	1	65	\$33	Uniform	0	66.41

* All unit values pulled from a lognormal distribution from Model 1, Table III in Dickie and Ulery are multiplied by 0.973811 to adjust for a difference in mean household income between the study participants and the general population. The unit values shown here have already been adjusted.

I.4.1 Non-Fatal Myocardial Infarctions (Heart Attacks)

See Appendix H, Section H.3.1 for an explanation of this valuation function.

Using age-specific annual lost earnings estimated by Cropper and Krupnick (1999), and a three percent discount rate, we estimated the following present discounted values in lost earnings over 5 years due to a heart attack (2015\$): \$13,301 for someone between the ages of 25 and 44, \$19,604 for someone between the ages of 45 and 54, and \$113,316 for someone between the ages of 55 and 65. The corresponding age-specific estimates of lost earnings using a seven percent discount rate are \$11,908, \$17,552, and \$101,451, respectively. Cropper and Krupnick do not provide lost earnings estimates for populations under 25 or over 65. As such we do not include lost earnings in the cost estimates for these age groups.

We have found three additional sources of estimates of the direct medical costs of a myocardial infarction (MI) in the literature:

- Wittels et al. (1990) estimated expected total medical costs of MI over 5 years to be \$51,211 (in 1986\$) for people who were admitted to the hospital and survived hospitalization. (There does not appear to be any discounting used.) Wittels et al. was used to value coronary heart disease in the 812 Retrospective Analysis of the Clean Air Act. Using the CPI-U for medical care, the Wittels estimate is \$187,530 in year 2015\$. This estimated cost is based on a medical cost model, which incorporated therapeutic options, projected outcomes and prices (using “knowledgeable cardiologists” as consultants). The model used medical data and medical decision algorithms to estimate the probabilities of certain events and/or medical procedures being used. The authors note that the average length of hospitalization for acute MI has decreased over time (from an average of 12.9 days in 1980 to an average of 11 days in 1983). Wittels et al. used 10 days as the average in their study. It is unclear how much further the length of stay (LOS) for MI may have decreased from 1983 to the present. The average LOS for ICD code 410 (MI) in the year-2000 AHQR HCUP database is 5.5 days. However, this may include patients who died in the hospital (not included among our non-fatal MI